

Bilal Delais

AI Engineer — Machine Learning & Generative AI

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Mobility: Paris · Toulouse · Bordeaux · Switzerland · International — Available from November 2026, fixed-term or permanent contract

PROFILE

AI & Data Science engineer (IMT Mines Alès, graduating November 2026) with two complementary areas of hands-on experience: **deep learning on industrial time series** (anomaly detection on rocket-engine test benches at ArianeGroup) and **generative AI deployed for a real client** (local LLM assistant, quote-analysis application). I turn vague business needs into tools that teams actually use.

EXPERIENCE

AI Engineer — Ochralab, architecture firm

Sept. – Nov. 2026 · Marrakech

- Designed and deployed a 100% on-premise AI assistant for a 3-person firm: open-source LLM (Gemma via Ollama) on a dedicated GPU workstation (RTX 3060 12 GB), heterogeneous sources (PDF plans and quotes, Excel, emails).
- Developed and rolled out a quote-comparison application built on the Claude API (Anthropic), installed on the firm's workstations.
- Reduced weekly time spent managing quotes by 50%.
- Gathered requirements directly from the principal architect; redesigned and deployed the firm's website.

AI / Data Science Engineer (final-year internship) — ArianeGroup

Mar. – Jul. 2026 · Vernon

- Multivariate anomaly-detection pipeline on time series from the Vulcain, Vinci and Prometheus engine test benches: Transformer-GAN, TranAD, LSTM Autoencoder (PyTorch).
- Addressed data scarcity (~300 test runs, fewer than 10 real anomalies) through synthetic anomaly generation and transfer learning; train/test split by test run, hyperparameter optimization with Optuna.

Machine Learning Developer — SpringLane, retail startup

Jun. 2025 – Mar. 2026 · Paris (remote)

- Stock-forecasting algorithm for an omnichannel retailer; optimized automatic replenishment using machine learning models.

TECHNICAL PROJECTS

- Framework-free multi-tool AI agent** — agent loop and tool calls implemented directly on the Anthropic API.
- Local RAG** — document search over a private corpus, fully offline (Ollama, ChromaDB, LangChain).
- Deep-learning correction of the SGP4 orbital propagator** — learning the residual error on TLE data (ISS, Hubble); MLP / LSTM / Transformer benchmark (team project).
- NoduLoCC 2026, medical AI challenge** — lung nodule detection on chest X-rays (RadDino, multitask classification + localization head).
- Event-based vision & self-supervised learning** — DETR on event cameras (DSEC-MOD), SimCLR, CLIP, DANN.

TECHNICAL SKILLS

- Generative AI:** LLMs (Claude API, Gemma / Ollama), RAG, agents and tool use, LangChain, ChromaDB, prompt engineering
- ML / deep learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Optuna, SHAP — time series, computer vision, NLP
- Data & languages:** Python, SQL, Pandas, NumPy
- Development & deployment:** Git, GitLab CI, Linux, Vercel, GitHub Pages
- Engineering:** SolidWorks

EDUCATION

Master's-level Engineering Degree (Diplôme d'ingénieur) — AI & Data Science

(2IA-IASD) — IMT Mines Alès

2023 – 2026 · graduating Nov. 2026

Classes préparatoires PSI (intensive undergraduate program in physics, engineering & mathematics) — Lycée Gustave Eiffel, Bordeaux

2020 – 2022

LANGUAGES & INTERESTS

French (native) · **English** (C1)

Running, weight training, basketball · aerospace